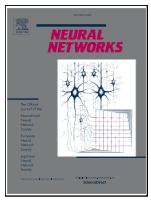
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Full Length Article



Deep multi-view graph clustering with multi-level contrastive guidance

Jiaxin Zhong [a], Bocheng Wang [b], Mulin Chen [a,*], Xuelong Li [c]

[a] School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, China
[b] School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an, China
[c] Institute of Artificial Intelligence (TeleAI), China Telecom Beijing, Beijing, 100033, China

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ABSTRACT

Deep graph-based multi-view clustering has gained prominence for modeling structural relationships and semantic patterns across views. Existing methods typically construct a graph followed by graph neural network processing, but suffer from two key limitations: (1) clustering performance is often constrained by defects in the graph construction; (2) most methods employing contrastive learning prioritize embedding discriminability while neglecting clustering adaptability. To overcome these challenges, we propose the Deep Multi-Level Contrastive Graph Clustering (DMCGC) approach. Specifically, feature-level contrastive learning is introduced to enhance cluster separability within each view and generate clustering-friendly attribute embeddings. Then, dual-scale graph learning is performed to extract view-shared local and global graph embeddings. Finally, attribute and graph embeddings are mapped into soft clustering assignments, regularized by cluster-level contrastive learning for structural consistency. Since the attribute embeddings convey the original characteristics and the graph embeddings are gradually optimized, the dependence on the initial graph is alleviated. Additionally, the clustering adaptability is also improved by the multi-level contrastive learning mechanism. Extensive experimental results on multi-view datasets demonstrate the competitive performance of DMCGC against state-of-the-art algorithms.

1. Introduction

In recent years, multi-view data arising from images, videos, audio, text, and other heterogeneous sources have been widely encountered in pattern recognition and data analysis. Unlike traditional single-view data, real-world samples are often characterized by multiple complementary views. For instance, a video clip can be described by its visual frames, audio signals, and textual subtitles, while an image can be represented by different descriptors such as color histograms, texture features, and deep visual features. As a fundamental unsupervised learning task, Multi-View Clustering (MVC) aims to integrate such complementary information to partition samples into semantically coherent groups (Guo et al., 2026; Zhao et al., 2023). This enables a wide range of applications, such as content understanding and generation (Zhang et al., 2024a), recommendation systems (Yu et al., 2023), and social network analysis (Tang et al., 2021). Numerous MVC strategies have been developed, including subspace learning-based methods (Liu et al., 2025; Yan et al., 2026), multiple kernel-based methods (Liu et al., 2026, 2020), deep learning-based methods (Wang et al., 2025; Zhao et al., 2021), among others (Fang et al., 2023).

Recently, graph-based methods have emerged as a powerful paradigm because of their ability to model structural relationships among samples. By representing samples as nodes and their pairwise relations as edges, these methods construct similarity graphs to characterize data manifolds and infer cluster structures (Wang et al., 2026). Deep learning has further advanced graph-based MVC by enabling the extraction of expressive and discriminative representations. Existing deep graph-based MVC methods not only leverage autoencoders for feature learning (Zheng et al., 2021), but also incorporate graph neural networks such as GCN (Kipf & Welling, 2016) to enhance structural modeling (Du et al., 2023).

Despite these advances, existing deep graph-based MVC methods still suffer from two limitations. (1) Limitations in graph construction. Some methods (e.g., NMvC-GCN Du et al., 2023) keep the graph fixed after initialization, making their performance highly sensitive to the quality of the initial graph. Other methods dynamically optimize graph structures (e.g., MLAN Nie et al., 2017b, DFMVC Zhao et al., 2024, and DOAGC Wen et al., 2024), but do not explicitly model complementary local neighborhoods and long-range global dependencies. (2) Insufficient clustering-oriented contrastive guidance. Existing methods often use contrastive learning to align embeddings across views, but mainly emphasize representation discriminability (e.g., MFLVC Xu et al., 2022b, GCFAgg Yan et al., 2023, DCD-DMVC Cui et al., 2024) while overlooking clustering adaptability, i.e., the compactness of intra-cluster

* Corresponding author.
E-mail addresses: 2025265592@mail.nwpu.edu.cn (J. Zhong), wangbocheng@mail.nwpu.edu.cn (B. Wang), chenmulin@nwpu.edu.cn (M. Chen), xuelong_li@ieee.org (X. Li).

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samples and separability of inter-cluster samples. Although discriminative embedding learning separates different samples, it may prevent the formation of a compact cluster structure and lead to suboptimal clustering results.

To address these challenges, we propose the Deep Multi-Level Contrastive Graph Clustering (DMCGC) framework. The main contributions of this paper are summarized as follows:

- A novel deep graph-based framework is proposed for multi-view clustering. It reduces initial graph dependency through dynamic graph learning and original attribute preservation. Meanwhile, clustering adaptability is improved by promoting more discriminative cluster assignments.
- A dual-scale graph learning mechanism is developed to jointly capture adaptive local neighborhoods and diffused global dependencies from evolving multi-view representations, providing complementary structural information for clustering.
- A multi-level contrastive learning strategy is designed to provide clustering-oriented guidance. Feature-level contrastive learning encourages the attribute embeddings to capture cluster-discriminative features, and cluster-level contrastive learning promotes consistent clustering across multi-view attributes and dual-scale graph representations.

The remainder of this paper is organized as follows. Section 2 reviews the most relevant related work. Section 3 introduces the proposed DMCGC framework, including feature-level contrastive learning, dual-scale graph learning, and cluster-level contrastive guidance. Section 4 presents the experimental settings and results. Finally, Section 5 concludes this paper. For clarity, Table 1 summarizes the main notations used in this paper.

2. Related works

This section reviews the most related works from three aspects, including multi-view clustering, graph-based deep multi-view clustering, and contrastive learning, and summarizes their key limitations that motivate our method.

2.1. Multi-view clustering

Multi-view clustering aims to group samples into clusters by leveraging the complementary and consistent information from different data views. Traditional MVC methods are typically categorized into subspace learning (Cao et al., 2015), graph-based clustering (Nie et al., 2017b), and matrix factorization (Liu et al., 2013). However, these traditional methods often rely on shallow architectures and linear embeddings, limiting their capacity to model complex and nonlinear relationships (Cai et al., 2024). Since DEC (Xie et al., 2016) first introduced autoencoders for clustering, deep learning has significantly advanced MVC. Deep MVC methods employ deep neural networks to learn nonlinear embeddings, allowing the extraction of complex patterns (Hu et al., 2020; Xu et al., 2021). Recent deep learning studies have further demonstrated the effectiveness of adaptive information integration and robust representation learning in complex data, including multi-granularity spatiotemporal fusion (Yu et al., 2025b), missing-value-aware multivariate modeling (Yu et al., 2025a), heterogeneous ensemble learning (Cheng & Liu, 2025), and meta-learning-enhanced attention modeling (Cheng et al., 2026). Related advances in domain generalization and cross-domain representation learning further highlight the importance of learning robust representations under heterogeneous data distributions (Zhang et al., 2025, 2026). Many advanced MVC frameworks have been proposed. For example, MFLVC (Xu et al., 2022b) adopts a fusion-free multi-level feature learning approach, while SDMVC (Xu et al., 2022a) leverages pseudo-labels and a unified target distribution to enhance feature discriminability. These approaches often outperform traditional MVC methods, especially on large-scale datasets.

Table 1
Summary of notations.

Notations	Meanings
N	Number of samples.
V	Number of views.
K	Number of clusters.
d_v	Feature dimension of the v-th view.
d_h	Dimension of the latent attribute/graph embeddings.
$\mathbf{X}^v \in \mathbb{R}^{N\times d_v}$	Data matrix of the v-th view.
$\mathbf{x}_i^v \in \mathbb{R}^{d_v}$	Feature vector of sample i in view v.
$\mathbf{H}^v \in \mathbb{R}^{N\times d_h}$	Attribute embedding matrix of view v.
$\mathbf{h}_i^v \in \mathbb{R}^{d_h}$	Attribute embedding vector of sample i in view v.
$\mathbf{H} \in \mathbb{R}^{N\times d_h}$	View-shared embedding matrix.
$\mathbf{S}^l, \mathbf{S}^0, \mathbf{S}^g \in \mathbb{R}^{N\times N}$	Local, initial global, and PPR-diffused global graphs.
$\mathbf{F}^l, \mathbf{F}^g \in \mathbb{R}^{N\times d_h}$	Local and global graph embedding matrices.
$\mathbf{Z}^v \in \mathbb{R}^{N\times K}$	Soft assignment matrix predicted from $\mathbf{H}^v$.
$\mathbf{P}^v \in \mathbb{R}^{N\times K}$	Enhanced assignment matrix of view v.
$\mathbf{p}_j^v \in \mathbb{R}^N$	Assignment vector of all samples to cluster j in view v.

2.2. Graph-based deep multi-view clustering

Since GCN was proposed, many deep MVC methods based on graph neural networks (GNNs) have achieved notable performance in clustering tasks (Fan et al., 2020; Huang et al., 2023). For example, MAGCN (Chen et al., 2020) introduces a multi-view attribute graph convolutional framework that uses attention mechanisms to reduce noise. DCRN (Liu et al., 2022) incorporates a graph propagation regularization mechanism to enhance the discriminability of nodes within the graph structure. However, these methods require explicit graph data as input, making them unsuitable for general clustering tasks without an original graph structure.

For real-world datasets without original graph structures, most deep graph-based MVC methods choose to first construct an initial graph. For example, CAGL (Wang et al., 2022) employs the k-nearest neighbors (KNN) algorithm to build attribute graphs for each view and uses graph encoders and attention networks to generate an attention-weighted fusion graph. DOAGC (Wen et al., 2024) adaptively reconstructs each input graph by combining feature-derived node correlations with the original graph structure, mainly addressing the heterophily issue in multi-view graph clustering. Related studies have also explored learnable graph operators for multi-modal feature fusion, such as LEGO (Ding et al., 2025), which models and integrates cross-modal feature relations in the graph space. However, these methods mainly focus on fixed graph construction or feature fusion, rather than clustering-oriented structural learning.

Some methods further attempt to dynamically update graph structures during training. For example, MLAN (Nie et al., 2017b) and DFMVC (Zhao et al., 2024) adaptively optimize sample relationships, but mainly focus on local neighborhood structures and lack explicit modeling of long-range dependencies. Other graph-based MVC methods also emphasize different structural aspects: MAGA (Bian et al., 2024) mainly exploits local graph information, while GCFAgg (Yan et al., 2023) focuses more on global cross-view aggregation. In contrast, DMCGC dynamically learns a view-shared local graph from evolving multi-view representations and further constructs a diffused global graph, allowing fine-grained neighborhood relations and long-range dependencies to provide complementary structural information for clustering.

2.3. Contrastive learning

Contrastive learning has emerged as a powerful strategy in MVC, addressing the challenge of learning discriminative representations across views (Hassani & Khasahmadi, 2020; Xu et al., 2024). By maximizing agreement between positive pairs and minimizing agreement between

negative pairs, contrastive learning improves the consistency and diversity of multi-view embeddings. For example, MFLVC (Xu et al., 2022b) adopts multi-level contrastive learning to enhance cross-view representation consistency, while DealMVC (Yang et al., 2023) introduces a dual contrastive calibration mechanism that aligns view-specific and view-fused features. However, these methods mainly impose contrastive constraints at the sample or representation level, which emphasize representation discriminability but do not directly coordinate heterogeneous information according to the final clustering distribution.

To better adapt contrastive learning to clustering, DMCGC introduces clustering-oriented guidance at both representation and assignment levels. Feature-level contrastive learning learns cluster-friendly attributes, while cluster-level alignment integrates attribute and dual-scale graph assignments for final clustering optimization.

3. Method

In this section, the proposed framework is presented. The pipeline of DMCGC is shown in Fig. 1. Feature-level contrastive learning first learns clustering-friendly attribute embeddings for each view. These embeddings are then dynamically fused to construct local and global graphs. Based on the two graphs, GCN and graph filtering generate dual-scale graph embeddings, which are finally mapped by SAN into soft cluster assignments. Cluster-level contrastive learning further aligns the graph- and attribute-based assignments.

Given a multi-view dataset $\mathcal{X} = \{\mathbf{X}^v \in \mathbb{R}^{N\times d_v}\}_{v=1}^{V}$, where N is the number of samples, V is the number of views, $\mathbf{X}^v = [\mathbf{x}_1^v, \mathbf{x}_2^v, \dots, \mathbf{x}_N^v]^T$ denotes the data matrix of the v-th view, $\mathbf{x}_i^v \in \mathbb{R}^{d_v}$ is the feature vector of sample i, and d_v is the dimension of the v-th view. The goal of DMCGC is to divide the samples into K clusters.

3.1. Pretraining network

In this part, to obtain discriminative features, our framework starts with view-specific autoencoders. The autoencoders are pretrained to facilitate convergence. The v-th view employs an encoder $f_{\theta_v}^v$ and a corresponding decoder $g_{\phi_v}^v$, with learnable parameters θ_v and ϕ_v, respectively. The embedding and reconstruction processes are represented by

$$\mathbf{h}_i^v = f_{\theta_v}^v(\mathbf{x}_i^v), \tag{1}$$

$$\hat{\mathbf{x}}_i^v = g_{\phi_v}^v(\mathbf{h}_i^v), \tag{2}$$

where $\mathbf{x}_i^v, \hat{\mathbf{x}}_i^v \in \mathbb{R}^{d_v}$ denote the input and reconstructed feature vectors, respectively, and $\mathbf{h}_i^v \in \mathbb{R}^{d_h}$ is the corresponding attribute embedding vector. For multiple views, the mean squared error (MSE) is used to pretrain all the autoencoders:

$$L_{pre} = \sum_{v=1}^{V}\sum_{i=1}^{N} \|\mathbf{x}_i^v - \hat{\mathbf{x}}_i^v\|_2^2. \tag{3}$$

While pretraining effectively optimizes data reconstruction, it neglects clustering structures. To address this limitation and derive clustering-oriented embeddings, we introduce a feature-level contrastive learning mechanism.

3.2. Feature-level contrastive learning

Based on the learned attribute embeddings, feature-level contrastive learning is proposed to enhance the cluster separability.

Typical contrastive learning treats embeddings of the same sample across different views as positive samples and those from different samples as negative ones. While this paradigm effectively enhances discriminability, it may increase the dispersion of all samples, potentially harming clustering performance. To derive clustering-friendly embeddings, we leverage cluster-center-based contrastive learning (Chen et al., 2024) as

$$l_{fl}(v_1, v_2) = \frac{1}{N}\sum_{i=1}^{N}\left(-\log\frac{e^{\theta(\mathbf{h}_i^{v_1}, \mathbf{h}_i^{v_2})/\tau}}{e^{\theta(\mathbf{h}_i^{v_1}, \mathbf{h}_i^{v_2})/\tau} + \sum_{j=1}^{K-1} e^{\theta(\mathbf{h}_i^{v_1}, \mathbf{q}_i^{j})/\tau}}\right), \tag{4}$$

where $\mathbf{h}_i^{v_1}, \mathbf{h}_i^{v_2} \in \mathbb{R}^{d_h}$ denote the embeddings of the i-th sample in views v_1 and v_2, respectively. At each training iteration, K-means is performed on the current $\mathbf{H}^{v_2} \in \mathbb{R}^{N\times d_h}$ to refresh the K cluster centroids, which are treated as non-differentiable references outside the computation graph. Gradients of $\mathcal{L}_{fl}$ are backpropagated through the attribute embeddings to update the encoders, and the centroids are recomputed in the next iteration. For each embedding $\mathbf{h}_i^{v_1}$, the positive pair is its corresponding embedding $\mathbf{h}_i^{v_2}$, while each negative centroid $\mathbf{q}_i^j \in \mathbb{R}^{d_h}$ is drawn from a different cluster. Accordingly, the negative centroid matrix $\mathbf{Q}_i \in \mathbb{R}^{(K-1)\times d_h}$ contains the $K-1$ irrelevant cluster centroids. Here, $\theta(\cdot,\cdot)$ computes the cosine similarity and τ is the temperature parameter.

Eq. (4) defines the contrastive loss between the two views v_1 and v_2. Accordingly, the feature-level contrastive loss across all views is defined as:

$$L_{fl} = \frac{1}{2}\sum_{v_1=1}^{V}\sum_{v_2\neq v_1}^{V} l_{fl}(v_1, v_2). \tag{5}$$

Eq. (5) aligns cross-view representations while repelling samples from irrelevant cluster centers, thereby enhancing cluster-oriented discriminability.

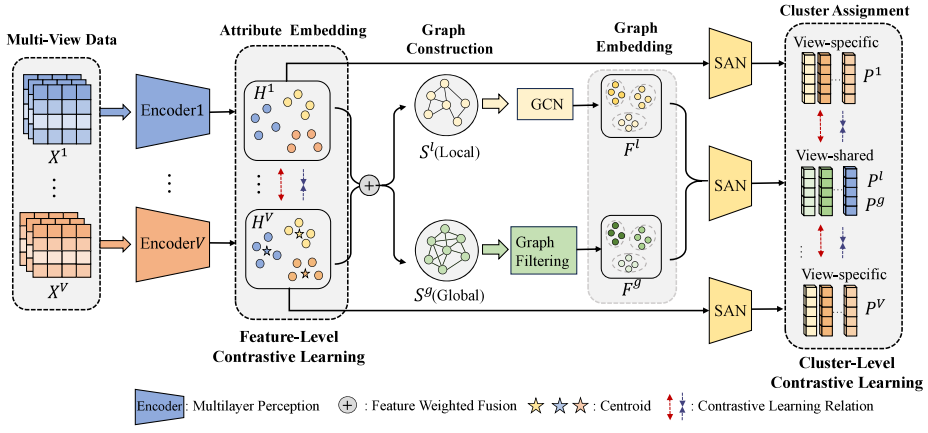


Fig. 1. The framework of DMCGC.

3.3. Dual-scale graph learning

The attribute embeddings obtained in Section 3.2 are clustering-friendly but fail to incorporate graph structural information. To provide clustering with complementary structural evidence, we construct two structural views with different roles: the local graph preserves fine-grained neighborhood relations shared across views, whereas the global graph complements it by capturing long-range dependencies that cannot be sufficiently covered by local neighborhoods.

3.3.1. Local graph learning

To construct a graph that integrates the neighbor relationships of all views, we formulate the local graph learning objective as

$$\min_{\mathbf{S}^l} \sum_v w_v \sum_{i,j} \left\| \mathbf{h}_i^v - \mathbf{h}_j^v \right\|_2^2 s_{ij}^l + \rho \|\mathbf{S}^l\|_F^2, \quad \text{s.t. } (\mathbf{s}_i^l)^T \mathbf{1} = 1,\ 0 \le s_{ij}^l \le 1, \tag{6}$$

where $\mathbf{S}^l \in \mathbb{R}^{N\times N}$ denotes the target local graph, $\mathbf{s}_i^l \in \mathbb{R}^N$ corresponds to the i-th column of $\mathbf{S}^l$, and s_{ij}^l is its scalar (i,j)-th element. Here, ρ represents the balancing parameter and w_v denotes the non-negative scalar weight of view v, which is learned and updated jointly with $\mathbf{S}^l$ during training. The second term is designed to prevent the trivial solution where $\mathbf{S}^l$ becomes the identity matrix. Accordingly, pairs with smaller Euclidean distances are assigned larger affinity weights. After multiple iterations of dynamically updating w_v and $\mathbf{S}^l$, the final view-shared local graph $\mathbf{S}^l$ is obtained. For large-scale datasets, we retain the k-nearest neighbors and store $\mathbf{S}^l$ sparsely, reducing graph storage from $\mathcal{O}(N^2)$ to approximately $\mathcal{O}(Nk)$.

3.3.2. Global graph learning

Although $\mathbf{S}^l$ preserves local neighborhood patterns, its intrinsic limitation in modeling global topological dependencies may induce structural fragmentation. To holistically model the latent data structures, we propose to learn a global graph.

Using view-specific attribute embeddings, a view-shared embedding $\mathbf{H} \in \mathbb{R}^{N\times d_h}$ is obtained through dynamic weighted fusion, and the initial global graph $\mathbf{S}^0 \in \mathbb{R}^{N\times N}$ is constructed according to Eq. (6) without the weight w_v. To capture the global dependencies and expand receptive fields without over-smoothing, Personalized PageRank (PPR) graph diffusion (Hassani & Khasahmadi, 2020) is applied to $\mathbf{S}^0$:

$$\mathbf{S}^g = \sum_{i=0}^{\infty} \delta(1-\delta)^i \left(\mathbf{D}_S^{-\frac{1}{2}} \mathbf{S}^0 \mathbf{D}_S^{-\frac{1}{2}} \right)^i, \tag{7}$$

where δ is the transport rate, and $\mathbf{D}_S \in \mathbb{R}^{N\times N}$ is the degree matrix of $\mathbf{S}^0$. Eq. (7) can be equivalently expressed in the following closed form:

$$\mathbf{S}^g = \delta \left[\mathbf{I} - (1-\delta) \mathbf{D}_S^{-\frac{1}{2}} \mathbf{S}^0 \mathbf{D}_S^{-\frac{1}{2}} \right]^{-1}, \tag{8}$$

where $\mathbf{I} \in \mathbb{R}^{N\times N}$ is the identity matrix and $\mathbf{S}^g \in \mathbb{R}^{N\times N}$ is the diffused global graph. Through Eq. (8), the resulting $\mathbf{S}^g$ captures multi-hop propagation effects, thereby capturing global structural information across views. For large-scale datasets, Eq. (8) is approximated by finite-step sparse PPR propagation, avoiding expensive matrix inversion while preserving global diffusion.

3.3.3. Graph embedding processing

By propagating the view-shared embedding $\mathbf{H}$ over $\mathbf{S}^l$ and $\mathbf{S}^g$, two types of view-shared graph embeddings are obtained.

Specifically, we first use GCN to process $\mathbf{S}^l$. Using $\mathbf{H}$ as the initial input and passing it through two layers of GCN, refined features are calculated as

$$\mathbf{F}^l = \mathrm{GCN}(\mathbf{H}, \mathbf{S}^l), \tag{9}$$

where $\mathbf{F}^l \in \mathbb{R}^{N\times d_h}$ is the local graph embedding matrix.

However, due to the layer limitations of GCN (Kipf & Welling, 2016) (e.g., over-smoothing), shallow architectures struggle to capture large receptive fields. To produce a smoother global representation, we apply graph filtering to $\mathbf{S}^g$:

$$\mathbf{F}^g = \left(\mathbf{I} - \frac{1}{2} \mathbf{L}_s \right)^t \mathbf{H}, \tag{10}$$

where $\mathbf{F}^g \in \mathbb{R}^{N\times d_h}$ is the global graph embedding matrix, $\mathbf{L}_s \in \mathbb{R}^{N\times N}$ is the normalized Laplacian matrix corresponding to $\mathbf{S}^g$, and t denotes the filtering order. By setting a larger t, each node aggregates information from more distant neighbors.

Consequently, $\mathbf{F}^l$ and $\mathbf{F}^g$ encode complementary local and global structural information, respectively. Rather than directly fusing them at the feature level, DMCGC subsequently projects them into the clustering space, where their complementary structural cues are explicitly coordinated through cluster-level contrastive learning.

3.4. Cluster assignment learning

After obtaining the attribute and graph embeddings, we convert all of them into cluster assignments for subsequent clustering-oriented optimization.

Given the view-shared graph embeddings $(\mathbf{F}^l, \mathbf{F}^g)$ and the view-specific attribute embeddings $[\mathbf{H}^1, \mathbf{H}^2, \ldots, \mathbf{H}^V]$, all with feature dimension d_h, the Shared Assignment Network (SAN), consisting of two linear layers and a softmax function, transforms them into soft clustering assignments. For a view-specific attribute embedding, the assignment is calculated as follows:

$$\mathbf{Z}^v = \mathrm{SAN}(\mathbf{H}^v), \tag{11}$$

where $\mathbf{Z}^v \in \mathbb{R}^{N\times K}$ denotes the assignment matrix of view v, and z_{ij}^v is the scalar probability that sample i belongs to cluster j.

For achieving more discriminative cluster assignments, we enhance the clustering probability distribution through the following strategies:

$$p_{ij}^v = \frac{\left(z_{ij}^v\right)^2 / \sum_{n=1}^N z_{nj}^v}{\sum_{k=1}^K \left(z_{ik}^v\right)^2 / \sum_{n=1}^N z_{nk}^v}. \tag{12}$$

Same as z_{ij}^v, p_{ij}^v is a scalar representing the probability that sample i in view v belongs to cluster j. Collecting these elements yields the enhanced assignment matrix $\mathbf{P}^v = [p_{ij}^v] \in \mathbb{R}^{N\times K}$. Accordingly, $\mathbf{p}_j^v = [p_{1j}^v, \ldots, p_{Nj}^v]^T \in \mathbb{R}^N$ denotes the assignment vector of all samples to cluster j in view v.

According to Eqs. (11) and (12), we obtain view-shared assignments $(\mathbf{P}^l, \mathbf{P}^g)$ and view-specific assignments $[\mathbf{P}^1, \mathbf{P}^2, \ldots, \mathbf{P}^V]$, where each assignment matrix is in $\mathbb{R}^{N\times K}$.

3.5. Cluster-level contrastive learning

Based on the clustering assignments, cluster-level contrastive learning is performed to achieve clustering alignment across different views.

To better adapt contrastive learning to cluster structures, we treat the assignment vector of all samples to a cluster as a unified unit. Assignments to the same cluster are aligned, while those to different clusters are separated. Unlike sample-based contrastive learning, this strategy enhances the inter-cluster discriminability while maintaining the consistency of clustering labels across views.

Cluster-level contrastive loss between two views is defined as

$$l_{cl}(v_1, v_2) = -\frac{1}{K} \sum_{k=1}^K \log \left(\frac{e^{s(\mathbf{p}_k^{v_1}, \mathbf{p}_k^{v_2})/\tau}}{T} \right),$$
$$T = \sum_{j=1, j\neq k}^K e^{s(\mathbf{p}_j^{v_1}, \mathbf{p}_k^{v_1})/\tau} + \sum_{j=1}^K e^{s(\mathbf{p}_j^{v_1}, \mathbf{p}_k^{v_2})/\tau}, \tag{13}$$
$$s(\mathbf{p}_k^{v_1}, \mathbf{p}_k^{v_2}) = \left(\mathbf{p}_k^{v_1}\right)^T \mathbf{p}_k^{v_2},$$

where $\mathbf{P}^{v_1}, \mathbf{P}^{v_2} \in \mathbb{R}^{N\times K}$ are the assignment matrices of the two views, $\mathbf{p}_k^{v_1}, \mathbf{p}_k^{v_2} \in \mathbb{R}^N$ are their k-th cluster-assignment vectors, and T and $s(\cdot,\cdot)$ are scalar quantities.

Given the view-shared graph-based assignments $\mathbf{P}^l$ and $\mathbf{P}^g$, we further integrate the original multi-view attribute information and dual-scale structural information in the clustering space. Specifically, the overall cluster-level contrastive loss is formulated as

$$L_{cl} = \beta \sum_{v=1}^{V} l_{cl}(v, l) + \alpha l_{cl}(l, g), \tag{14}$$

where β and α are scalar trade-off coefficients.

The first term aligns the local graph assignment $\mathbf{P}^l$ with the view-specific attribute assignments $\{\mathbf{P}^v\}_{v=1}^V$, preserving complementary attribute information in graph-based clustering. The second term aligns $\mathbf{P}^l$ with the global graph assignment $\mathbf{P}^g$, promoting consistency between local neighborhood and long-range structural information. Thus, L_{cl} integrates multi-view attributes and dual-scale structures at the assignment level for joint clustering optimization.

Finally, to balance cluster sizes and prevent collapse into a single cluster, we impose an entropy-based regularization:

$$L_{en} = \sum_{v=1}^{V} \sum_{j=1}^{K} g_j^v \log g_j^v + \sum_{j=1}^{K} g_j^l \log g_j^l, \tag{15}$$

where g_j^v is the scalar average assignment probability defined as $g_j^v = \frac{1}{N} \sum_{i=1}^{N} p_{ij}^v$. By minimizing L_{en}, we maximize the entropy of the average assignment distributions g_j^v and g_j^l, thus encouraging the assignments to spread evenly across all clusters and avoiding the dominance of a single cluster.

3.6. Overall clustering model

The overall loss function of DMCGC is

$$L = L_{pre} + \lambda L_{fl} + L_{cl} + \gamma L_{en}, \tag{16}$$

where L, L_{pre}, L_{fl}, L_{cl}, and L_{en} are scalar losses, and λ and γ are scalar trade-off parameters.

Let $\mathbf{p}_i^l = [p_{i1}^l, \ldots, p_{iK}^l] \in \mathbb{R}^K$ be the i-th row of $\mathbf{P}^l$, and let p_{ij}^l denote its j-th scalar element. After network training is completed, the clustering label for sample i can be predicted using the cluster assignment vector $\mathbf{p}_i^l$:

$$y_i = \arg\max_j \left(p_{ij}^l \right). \tag{17}$$

DMCGC first uses L_{pre} to pretrain autoencoders, and then trains the entire network with L. Training is stopped after a predefined number of epochs. The optimizer of DMCGC is Adam. The complete flow of DMCGC is displayed in Algorithm 1.

Algorithm 1 DMCGC.

Input: Data matrices $\{\mathbf{X}^v\}_{v=1}^V$, with N samples and K clusters, epochs *epochs*, parameters α, β, γ and λ.
Initialize $\{\theta_v\}_{v=1}^V$, $\{\phi_v\}_{v=1}^V$ by minimizing L_{pre} in Eq. (3);
for $i = 1$ **to** *epochs* **do**
  Compute $\{\mathbf{H}^v\}_{v=1}^V$ via Eq. (1) and refresh the K-means centroids;
  Compute $\mathbf{H}$;
  Compute $\mathbf{S}^l$ by solving Eq. (6);
  Compute $\mathbf{S}^g$ via Eq. (8) (finite-step sparse PPR for large-scale data);
  Obtain $\mathbf{F}^l$ and $\mathbf{F}^g$ by GCN and graph filtering.
  Compute $\mathbf{P}^l$, $\mathbf{P}^g$ and $\{\mathbf{P}^v\}_{v=1}^V$ via Eqs. (11) and (12);
  Compute L_{pre}, L_{fl}, L_{cl} and L_{en}, respectively;
  Update model by minimizing Eq. (16) with Adam.
end for
Obtain clustering labels by Eq. (17);
Output: The predicted labels $\{y_i\}_{i=1}^N$.

Table 2
Datasets used in the experiment.

Dataset	Samples	Classes	Views	Dimensions
Handwritten	2000	10	6	240, 76, 216, 47, 64, 6
COIL-20	1440	20	3	1024, 3304, 6750
Prokaryotic	551	4	3	438, 3, 393
Hdigit	10,000	10	2	784, 256
RGB-D	1449	13	2	2048, 600
Reuters	18758	6	5	21531, 24892, 34251, 15506, 11547
Caltech-2V	1400	7	2	40, 254
Caltech-3V	1400	7	3	40, 254, 1984
Caltech-4V	1400	7	4	40, 254, 1984, 512
Caltech-5V	1400	7	5	40, 254, 1984, 512, 928

3.7. Computational complexity

For the dense formulation, graph construction and propagation require $\mathcal{O}(VN^2 d_h)$ and $\mathcal{O}(N^2 d_h)$ time, respectively, with $\mathcal{O}(N^2)$ graph storage for fixed V. Centroid updates, feature-level contrast, and cluster-level contrast cost $\mathcal{O}(V I_{\mathrm{km}} N K d_h)$, $\mathcal{O}(V^2 N K d_h)$, and $\mathcal{O}((V+1)NK^2)$, respectively, where I_{km} denotes the K-means iteration count. The V^2 factor arises from pairwise comparisons across views. Although explicit inversion in Eq. (8) requires $\mathcal{O}(N^3)$ time, replacing it with finite-step PPR yields an overall per-training-iteration cost of $\mathcal{O}(N^2)$ when model dimensions and internal iteration counts are fixed. For large graphs, retaining k neighbors and applying R-step sparse PPR directly to features reduces graph storage to $\mathcal{O}(Nk)$ and propagation cost to $\mathcal{O}(RNkd_h)$ for a fixed filtering order; exhaustive neighbor search remains quadratic in N.

4. Experiment

This section evaluates DMCGC on several public multi-view datasets. It first introduces the experimental setup (datasets, metrics, baselines, and parameter settings), then reports performance comparisons under different view numbers, and finally provides ablation and parameter sensitivity analyses to validate the effectiveness and robustness of each component.

4.1. Experimental setup

4.1.1. Datasets

We evaluate DMCGC on seven publicly available multi-view datasets. These datasets include: Handwritten (Asuncion et al., 2007), COIL-20 (Nene et al., 1996), Prokaryotic (Brbić et al., 2016), Hdigit (Chen et al., 2022), RGB-D (Kong et al., 2014), Reuters (Amini et al., 2009), and Caltech-5V (Fei-Fei et al., 2004). Detailed information on these datasets is provided in Table 2.

4.1.2. Evaluation metrics

To assess the clustering performance of all competing algorithms, three metrics are used: Clustering Accuracy (ACC), Normalized Mutual Information (NMI), and Purity (PUR) (Kong et al., 2011). Higher values indicate better clustering performance for all three metrics. The results are reported as the mean ± standard deviation over 10 independent runs.

4.1.3. Comparison methods

Twelve representative MVC methods are used for comparison. Traditional methods are MLAN (Nie et al., 2017a) and BSVC (Chen et al., 2023). Deep algorithms include MFLVC (Xu et al., 2022b), SiMVC (Trosten et al., 2021), CoMVC (Trosten et al., 2021), DEMVC (Xu et al., 2021), SGFCC (Shu et al., 2024), MAGA (Bian et al., 2024), GCFAgg (Yan et al., 2023), CVCL (Chen et al., 2023), DDMVC (Xu et al., 2025), SSLNMVC (Yan et al., 2025).

Table 3
Results of all methods on datasets (Part I). Results are reported as mean ± standard deviation (%).

Dataset	Handwritten			COIL-20			Prokaryotic		
Method	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
BSVC	75.48±0.00	74.21±0.00	75.48±0.00	81.20±0.00	84.71±0.00	81.32±0.00	52.27±0.00	24.43±0.00	58.26±0.00
MLAN	97.35±0.00	94.00±0.00	97.35±0.00	82.28±0.00	86.94±0.00	82.74±0.00	56.62±0.00	24.01±0.00	58.26±0.00
MFLVC	86.55±1.24	85.98±0.77	86.55±1.24	73.19±1.72	81.43±1.21	75.07±1.58	43.01±4.22	22.16±2.94	59.89±3.22
SiMVC	79.63±2.99	77.82±1.18	79.63±2.99	70.12±2.61	82.86±1.52	73.52±2.46	50.09±3.79	19.45±2.58	60.98±3.31
CoMVC	83.37±3.38	85.98±1.09	83.37±3.38	78.52±2.38	86.06±1.38	78.52±2.22	41.38±4.23	18.83±2.84	66.97±3.03
DEMVC	67.69±5.94	70.61±3.04	67.69±5.94	69.40±4.36	77.90±2.83	70.10±4.06	51.23±4.93	30.28±3.42	67.26±3.83
SGFCC	63.25±4.54	62.69±3.29	63.70±4.18	44.17±4.72	60.40±3.25	44.17±4.46	57.17±4.41	26.80±3.11	68.06±3.66
MAGA	92.20±0.78	84.66±1.87	92.20±0.78	51.39±2.12	63.12±1.73	53.68±1.93	58.08±2.73	25.41±2.22	59.35±2.44
GCFAgg	84.95±1.47	77.71±3.07	84.95±1.47	63.40±1.72	71.67±1.54	67.01±1.64	62.25±2.56	**37.78±2.33**	**73.14±2.32**
CVCL	97.35±0.64	94.05±0.57	97.35±0.64	84.65±0.89	**88.89±0.61**	85.07±0.82	56.19±4.14	25.05±3.29	57.01±3.75
DDMVC	90.50±0.36	84.60±0.62	90.50±0.36	54.51±1.46	61.28±1.23	55.56±1.39	64.07±2.69	16.01±2.14	64.25±2.56
SSLNMVC	85.61±1.11	78.26±1.98	85.61±1.11	64.30±1.55	72.15±1.38	65.23±1.57	66.48±2.53	37.14±2.26	67.31±2.42
DMCGC (ours)	**97.55±0.67**	**94.41±0.94**	**97.55±0.67**	**84.93±0.98**	88.66±1.68	**85.12±1.22**	**68.60±2.77**	36.33±1.97	68.60±2.77

Table 4
Results of all methods on datasets (Part II). Results are reported as mean ± standard deviation (%).

Dataset	Hdigit			RGB-D			Reuters		
Method	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
BSVC	66.27±0.00	71.46±0.00	70.53±0.00	10.69±0.00	33.04±0.00	40.72±0.00	33.37±0.00	12.23±0.00	34.28±0.00
MLAN	56.13±0.00	63.92±0.00	60.58±0.00	27.26±0.00	36.84±0.00	43.75±0.00	30.26±0.00	11.42±0.00	34.12±0.00
MFLVC	94.42±0.59	87.50±0.46	94.42±0.55	42.17±2.58	25.32±1.95	46.10±2.32	47.19±0.83	32.51±0.64	58.03±4.31
SiMVC	78.54±1.24	67.05±0.91	78.54±1.24	39.60±3.18	35.60±2.32	41.26±2.84	44.02±2.03	26.58±1.21	52.19±3.02
CoMVC	90.32±0.81	87.13±0.79	90.32±0.81	41.30±2.64	**40.50±2.06**	48.50±2.45	49.39±0.92	36.25±1.14	62.29±2.19
DEMVC	37.38±1.76	32.55±1.42	48.16±1.68	36.05±3.75	22.78±2.58	38.16±3.14	48.69±1.04	34.56±1.05	60.47±2.13
SGFCC	97.18±0.57	92.67±0.45	97.18±0.57	40.92±3.33	22.21±2.45	44.93±3.27	42.57±2.94	26.37±1.85	48.92±2.34
MAGA	91.41±0.75	80.27±0.62	91.41±0.75	35.96±2.28	17.52±1.75	38.37±2.06	50.76±2.29	39.69±2.04	65.35±2.77
GCFAgg	97.44±0.41	93.05±0.48	97.44±0.41	28.92±1.95	22.98±1.66	45.55±1.83	52.75±1.86	40.39±1.45	65.75±2.45
CVCL	99.34±0.13	98.01±0.12	99.34±0.13	40.08±1.97	32.39±1.63	50.88±2.02	45.16±2.08	29.48±5.17	56.79±3.24
DDMVC	95.34±0.51	88.54±0.59	98.34±0.38	36.37±1.77	18.60±1.51	36.37±1.77	43.37±3.25	28.68±2.45	50.11±2.59
SSLNMVC	99.00±0.31	96.97±0.37	99.00±0.31	31.33±1.83	23.22±1.57	45.96±1.89	46.58±2.39	34.76±1.87	59.35±2.51
DMCGC (ours)	**99.62±0.15**	**98.80±0.25**	**99.62±0.15**	**48.86±1.55**	37.25±1.34	**53.21±1.51**	**54.03±2.39**	**41.59±3.69**	**67.36±2.64**

4.1.4. Parameter settings

DMCGC is implemented in PyTorch and optimized using Adam. The propagation rate δ and temperature parameter τ are fixed to 0.2 and 0.5, respectively. The neighborhood size k is selected from $\{2, 3, 5\}$ and the graph filtering order t from $\{3, 5\}$. The learning rate is selected from $\{1 \times 10^{-4}, 3 \times 10^{-4}, 5 \times 10^{-4}\}$, while α, β, λ, and γ are selected from $\{0.00005, 0.0002, 0.0005, 0.002, 0.004, 0.01, 0.02\}$. The autoencoders contain 3–5 hidden layers, and the SAN consists of two linear layers followed by softmax. Pre-training and joint training use 320–900 and 30–600 epochs, respectively, depending on the dataset. Once selected, all configurations are fixed across independent runs. Ground-truth labels are used only for final evaluation.

4.2. Performance comparison

The clustering performance of all competing methods on seven datasets is summarized in Tables 3–5, with the highest and second-highest metric values highlighted in bold and underlined, respectively. As demonstrated in these tables, DMCGC achieves competitive results across the benchmarks and obtains the best performance on multiple metrics. Key observations are outlined as follows: 1) Traditional graph clustering methods (e.g., BSVC, MLAN) perform well on image datasets like Handwritten and COIL-20 but struggle with heterogeneous or relatively large datasets such as RGB-D and Reuters. Deep methods generally perform better on these datasets, benefiting from stronger semantic extraction. 2) Compared to existing contrastive learning methods (e.g., MFLVC, CoMVC, SGFCC, MAGA, and GCFAgg), our approach demonstrates superior performance using cluster-based contrastive learning. Moreover, by integrating graph-based embeddings, our framework also achieves competitive or superior results compared with CVCL, DDMVC, and SSLNMVC. In particular, DMCGC achieves the best ACC and NMI on Reuters, demonstrating its effectiveness on a relatively large-scale dataset. 3) While both MAGA and GCFAgg employ graph frameworks, the absence of a global graph in MAGA and the lack of local neighbor graphs in GCFAgg limit their ability to adapt to heterogeneous data. In comparison, our method demonstrates enhanced robustness by incorporating more diverse graph structures.

To further test the impact of different view numbers on the final clustering results, we evaluated the performance of all comparison methods on the Caltech dataset with varying numbers of views. The results are presented in Table 5. Although the ACC and NMI values of DMCGC are slightly lower than those of several competing methods in the two-view dataset, our performance surpasses all baselines as views increase. This trend highlights our method's ability to integrate complementary information while suppressing view-specific noise. This robustness stems from our method's simultaneous learning of view-consistent and view-specific clustering assignments during cluster-level contrastive learning.

4.3. Sensitivity to initial graph quality

To directly evaluate the sensitivity to initial graph quality, we randomly rewire 0%, 20%, and 40% of the existing edges while preserving the graph sparsity. The perturbed graph is used only for initialization, after which DMCGC follows its original dynamic graph learning procedure without further perturbation. We compare DMCGC with the representative fixed-graph method DFP-GNN (Xiao et al., 2023) and adaptive-graph (MLAN and MAGA) baselines under the same perturbation ratios. As shown in Fig. 2, DMCGC exhibits smaller performance degradation on both datasets. From 0% to 40% perturbation, its ACC decreases by

Table 5
Results of all methods on Caltech with different views. "-XV" represents that there are X views. Results are reported as mean ± standard deviation (%).

Dataset	Caltech-2V			Caltech-3V			Caltech-4V			Caltech-5V		
Method	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
MLAN	48.14±0.00	43.81±0.00	48.50±0.00	29.93±0.00	31.46±0.00	31.57±0.00	31.36±0.00	32.03±0.00	32.07±0.00	31.36±0.00	31.82±0.00	32.07±0.00
MFLVC	60.60±1.65	52.80±1.37	61.60±1.57	63.12±1.37	56.63±1.16	63.92±1.33	73.32±1.17	65.23±0.97	73.42±1.19	80.42±1.02	70.32±0.85	80.43±0.97
SiMVC	50.83±2.34	47.15±1.93	55.73±2.23	56.92±2.12	49.53±1.71	59.12±1.92	61.93±1.86	53.62±1.59	63.03±1.76	71.93±1.75	67.71±1.31	72.92±1.52
CoMVC	46.63±2.28	42.62±1.86	52.72±2.07	54.13±1.92	50.43±1.69	58.42±1.89	56.83±1.79	56.92±1.44	64.63±1.67	70.03±1.51	68.71±1.22	74.62±1.41
DEMVC	46.00±3.56	36.66±2.75	49.50±3.21	49.29±3.15	45.04±2.43	51.86±2.95	53.36±2.98	41.36±2.24	53.36±2.75	49.86±2.85	38.45±2.18	52.07±2.68
SGFCC	64.51±1.67	53.28±1.34	64.77±1.59	69.21±1.35	62.52±1.14	70.43±1.39	84.93±1.13	74.17±0.91	85.00±1.04	87.07±0.94	77.14±0.81	87.07±0.94
MAGA	58.79±1.54	50.21±1.37	60.29±1.57	63.36±1.34	58.13±1.14	66.14±1.22	74.00±1.05	63.49±0.93	74.00±1.05	86.07±0.98	77.31±0.72	86.07±0.88
GCFAgg	63.71±1.74	50.74±1.43	63.93±1.61	64.00±1.45	53.45±1.28	65.29±1.36	73.43±1.21	66.10±1.08	73.43±1.16	83.36±1.06	73.31±0.97	83.36±1.06
CVCL	58.10±1.42	51.86±1.21	62.30±1.45	72.50±1.23	66.50±1.05	72.50±1.16	84.20±1.04	75.43±0.96	84.20±1.04	81.50±1.07	78.35±0.89	82.10±0.91
DDMVC	**66.64±1.25**	**61.18±1.18**	66.64±1.25	69.71±1.01	60.95±0.94	70.57±1.03	77.29±0.97	72.74±0.82	77.29±0.97	82.07±0.89	76.26±0.73	82.07±0.89
SSLNMVC	62.93±1.22	53.52±1.02	63.21±1.18	65.07±1.01	57.43±0.97	65.21±0.99	84.57±0.87	74.22±0.73	84.57±0.87	87.86±0.76	79.02±0.62	87.86±0.76
DMCGC (ours)	65.43±2.24	54.35±1.93	**67.14±1.76**	**83.93±0.84**	**73.60±0.78**	**83.87±0.83**	**87.71±0.76**	**78.55±0.68**	**87.90±0.75**	**88.71±1.04**	**79.19±0.68**	**88.85±0.74**

Table 6
Ablation study of different global graph construction strategies. Results are reported as mean ± standard deviation (%).

Dataset	Metric	DMCGC-Sim	DMCGC-KNN	DMCGC
Handwritten	ACC	91.80±1.15	93.32±0.88	**97.55±0.67**
	NMI	88.45±0.44	91.17±1.21	**94.41±0.94**
	PUR	92.06±0.95	93.32±0.88	**97.55±0.67**
COIL-20	ACC	76.94±2.42	80.19±1.24	**84.93±0.98**
	NMI	83.85±1.52	84.91±0.95	**88.66±1.68**
	PUR	78.48±1.92	80.77±1.53	**85.12±1.22**
RGB-D	ACC	45.60±2.05	46.72±1.24	**48.86±1.55**
	NMI	33.18±1.83	34.64±1.62	**37.25±1.34**
	PUR	50.13±1.99	51.28±1.93	**53.21±1.51**

nearly 9 and 13 percentage points on COIL-20 and Caltech-5V, respectively, which are smaller than the decreases observed for the compared methods. These results indicate that dynamic graph learning alleviates the influence of imperfect graph initialization.

4.4. Visualization analysis

To provide an intuitive understanding of the learning process, we visualize the local graph embeddings learned by DMCGC on Caltech-5V at different training stages, as shown in Fig. 3. Different colors denote ground-truth classes only for visualization and are not used during training.

As training proceeds, the initially mixed representations gradually evolve into more compact and well-separated structures. This evolution illustrates the progressive refinement of the learned graph representations, while the increasing correspondence between the clusters and reference classes qualitatively reflects the improvement of clustering assignments. These results provide an intuitive view of the learned embeddings, graph-representation evolution, and clustering structure during optimization.

4.5. Ablation studies

This section first examines the influence of pre-training and the effectiveness of PPR-based global graph diffusion. It then demonstrates the essential role of DMCGC's core loss modules, and finally establishes the effectiveness of the clustering-oriented mechanism.

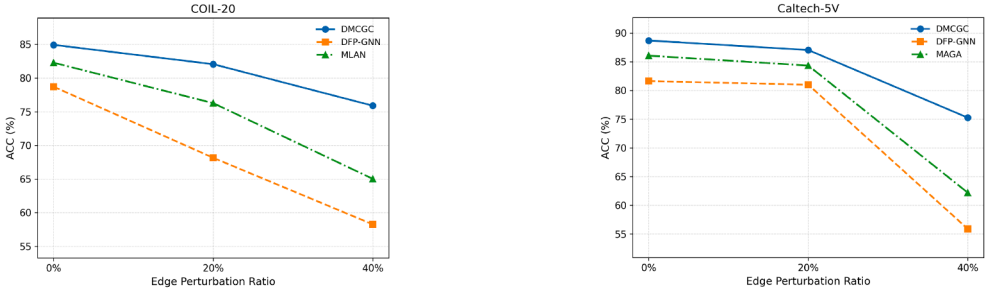
4.5.1. Influence of pre-training

To evaluate the effectiveness of DMCGC's two-stage training strategy, we denote the variant without pre-training as DMCGC-P. We compare DMCGC with DMCGC-P on four datasets, and the results are visualized in Fig. 4. Pre-training substantially improves clustering performance, especially on RGB-D, where ACC, NMI, and PUR increase by 19.53, 27.94, and 20.50 percentage points, respectively.

4.5.2. Effectiveness of PPR-based global diffusion

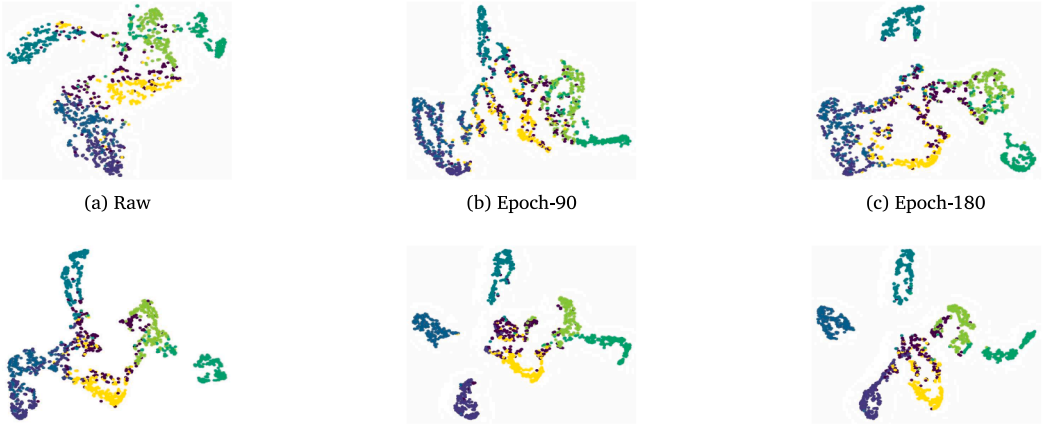
To evaluate the contribution of PPR-based global diffusion, we compare DMCGC with two variants that replace the PPR-diffused global graph with the original similarity graph (DMCGC-Sim) and a conventional KNN graph (DMCGC-KNN), respectively. All other model components and training settings are kept unchanged for a fair comparison. The results are reported in Table 6.

As shown in Table 6, DMCGC consistently achieves better mean performance than DMCGC-Sim and DMCGC-KNN across the three datasets. Compared with directly using the original similarity graph or a local KNN graph, PPR diffusion propagates structural information beyond immediate neighborhoods and therefore provides more comprehensive global dependencies for clustering. The improvements are particularly evident on Handwritten and COIL-20, demonstrating the effectiveness of PPR-based global structural modeling.



(a) COIL-20 (b) Caltech-5V

Fig. 2. Clustering performance under different perturbation ratios of the initial graph.



(a) Raw (b) Epoch-90 (c) Epoch-180

(d) Epoch-270 (e) Epoch-360 (f) Final

Fig. 3. Visualization of the local graph embeddings learned by DMCGC on Caltech-5V.

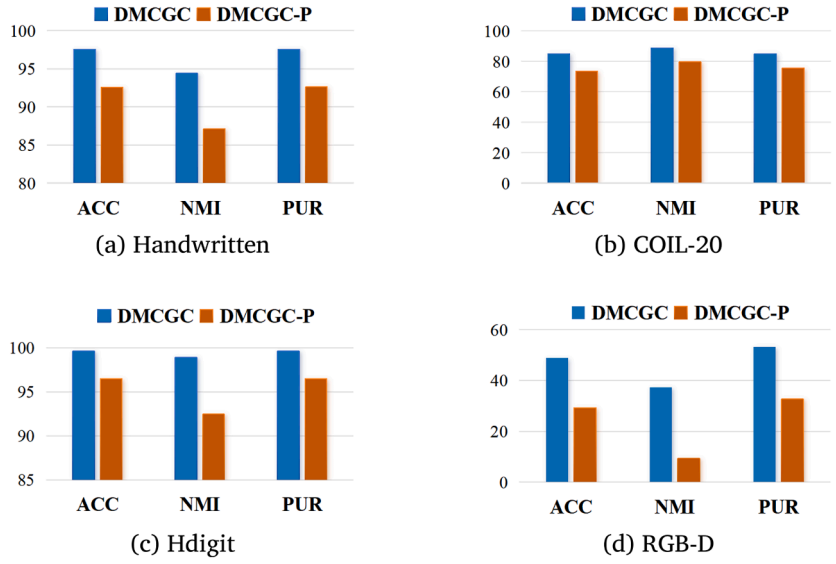


Fig. 4. Influence of pre-training on results.

4.5.3. Necessity of core loss modules

According to Eq. (16), the overall loss consists of four core components: reconstruction loss, feature-level and cluster-level contrastive losses, and clustering balance regularization. To validate their necessity, we perform ablation studies by isolating each module. Specifically, based on Eq. (14), we decompose the cluster-level contrastive loss L_{cl} into L_{cl}^1 (i.e., the contrastive loss between the local fusion view and each original view, corresponding to the first term of L_{cl}) and L_{cl}^2 (i.e., the contrastive loss between the local fusion view and the global fusion view, the second term of L_{cl}). Removing $\mathcal{L}_{cl}^1$ evaluates the interaction between view-specific attributes and the local graph representation, while removing $\mathcal{L}_{cl}^2$ eliminates the contribution of the global structural branch to clustering optimization, thereby allowing the complementary roles of local and global structures to be examined separately. By removing L_{cl}^1, L_{cl}^2, L_{fl}, and L_{en}, we analyze their individual contributions, with results detailed in Table 7 and Table 8.

Table 7
Ablation results concerning the main components of the proposed DMCGC method on Handwritten, COIL-20, and Prokaryotic. Results are reported as mean ± standard deviation (%).

L_{cl}^1	L_{cl}^2	L_{fl}	L_{en}	Handwritten			COIL-20			Prokaryotic		
				ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
	✓	✓	✓	92.05±1.55	89.35±1.01	92.05±1.55	83.33±1.29	87.32±0.92	83.82±1.19	59.53±3.07	15.68±2.19	59.53±2.84
✓		✓	✓	96.70±0.87	93.25±0.68	96.70±0.87	77.99±1.79	83.24±1.33	79.86±1.66	68.42±2.37	33.81±2.09	68.42±2.37
✓	✓		✓	91.25±1.64	89.48±1.23	91.25±1.64	81.11±1.46	86.14±1.06	81.32±1.31	67.33±2.43	30.47±2.13	67.33±2.25
✓	✓	✓		68.70±4.07	84.42±1.84	68.85±3.97	83.19±1.23	87.16±0.95	83.75±1.14	66.24±2.76	30.26±2.35	66.24±2.63
✓	✓	✓	✓	**97.55±0.67**	**94.41±0.94**	**97.55±0.67**	**84.93±0.98**	**88.66±1.68**	**85.12±1.22**	**68.60±2.77**	**36.33±1.97**	**68.60±2.77**

Table 8
Ablation results concerning the main components of the proposed DMCGC method on Hdigit, RGB-D, and Caltech-5V. Results are reported as mean ± standard deviation (%).

L_{cl}^1	L_{cl}^2	L_{fl}	L_{en}	Hdigit			RGB-D			Caltech-5V		
				ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
	✓	✓	✓	90.02±1.06	86.79±0.84	90.02±1.06	44.10±2.02	31.46±1.79	46.72±2.07	85.93±1.04	75.17±0.82	85.93±1.04
✓		✓	✓	98.98±0.26	97.10±0.39	98.98±0.26	45.89±1.87	36.28±1.43	52.10±1.76	83.07±1.31	70.37±1.07	83.07±1.23
✓	✓		✓	98.90±0.21	96.94±0.32	98.90±0.21	46.58±1.77	37.20±1.36	**53.62±1.71**	86.71±0.91	77.39±0.79	86.71±0.91
✓	✓	✓		79.56±1.47	91.86±0.76	79.56±1.47	38.85±2.65	23.81±2.04	41.48±2.58	61.79±2.19	64.27±1.74	61.79±2.07
✓	✓	✓	✓	**99.62±0.15**	**98.80±0.25**	**99.62±0.15**	**48.86±1.55**	**37.25±1.34**	53.21±1.51	**88.71±1.04**	**79.19±0.68**	**88.85±0.74**

Table 9
Performance comparison of three variants on various datasets. Results are reported as mean ± standard deviation (%).

Dataset	Metric	DMCGC-F	DMCGC-C	DMCGC-B	DMCGC
Handwritten	ACC	93.42±1.02	80.09±2.65	78.75±3.05	**97.55±0.67**
	NMI	93.17±0.85	78.65±2.12	72.45±2.59	**94.41±0.94**
	PUR	93.42±1.02	80.09±2.65	78.75±3.05	**97.55±0.67**
COIL-20	ACC	82.46±1.12	78.97±1.84	78.75±2.03	**84.93±0.98**
	NMI	85.17±0.93	85.02±1.29	84.23±1.41	**88.66±1.68**
	PUR	83.65±1.16	79.66±1.73	79.03±1.93	**85.12±1.22**
Prokaryotic	ACC	63.79±2.52	52.39±3.42	51.18±3.84	**68.60±2.77**
	NMI	24.32±2.27	12.25±1.92	10.49±1.72	**36.33±1.97**
	PUR	63.79±2.46	54.18±3.24	60.25±3.06	**68.60±2.77**
Hdigit	ACC	99.10±0.25	92.17±0.95	91.65±1.06	**99.62±0.15**
	NMI	98.04±0.23	86.42±0.74	84.68±0.83	**98.80±0.25**
	PUR	99.10±0.25	92.17±0.95	91.65±1.06	**99.62±0.15**
RGB-D	ACC	47.43±1.63	45.91±1.96	45.62±2.26	**48.86±1.55**
	NMI	36.81±1.42	35.46±1.65	34.21±1.91	**37.25±1.34**
	PUR	52.69±1.64	51.37±1.85	50.93±2.03	**53.21±1.51**
Caltech-5V	ACC	82.76±1.11	73.95±1.89	71.43±2.24	**88.71±1.04**
	NMI	73.45±0.91	70.89±1.46	67.23±1.67	**79.19±0.68**
	PUR	84.42±1.01	73.95±1.89	74.36±1.77	**88.85±0.74**

Table 10
Computational efficiency comparison on Reuters.

Method	Reference Complexity	Training Time (s)	Peak GPU Memory (GB)
MFLVC	$\mathcal{O}(N^2)$	615.00	0.52
MAGA	$\mathcal{O}(N^2)$	1964.82	0.53
GCFAgg	$\mathcal{O}(N^2)$	351.57	1.62
SSLNMVC	$\mathcal{O}(N^2)$	1030.85	0.74
DMCGC (ours)	$\mathcal{O}(N^2)$	414.84	1.45

N denotes the number of samples. Reference complexities assume full-data pairwise evaluation with fixed view numbers, dimensions, cluster numbers, and internal iteration counts; they do not describe actual mini-batch training costs. DMCGC uses finite-step PPR without explicit matrix inversion.

Table 11
Sensitivity analysis of the graph filtering order t on COIL-20 and Caltech-5V. Results are reported as mean ± standard deviation (%).

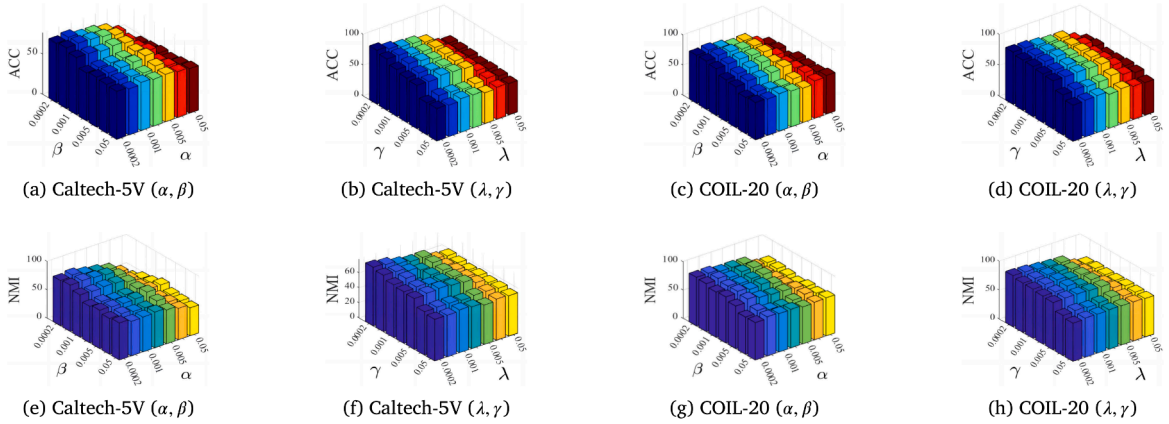
t	COIL-20		Caltech-5V	
	ACC	NMI	ACC	NMI
1	80.73 ± 2.29	86.54 ± 1.43	83.84 ± 1.39	76.52 ± 0.96
3	84.08 ± 1.05	87.71 ± 1.39	87.56 ± 1.18	78.39 ± 0.77
5	**84.93 ± 0.98**	**88.66 ± 1.68**	**88.71 ± 1.04**	**79.19 ± 0.68**
7	83.75 ± 1.17	87.54 ± 1.45	86.72 ± 1.23	77.50 ± 0.88
9	81.30 ± 1.94	86.88 ± 1.56	83.64 ± 1.56	75.48 ± 1.11

DMCGC with all components achieves the best mean performance on most metrics across all datasets, except for PUR on RGB-D, where removing L_{fl} yields 53.62% compared with 53.21% for the full model. Notably, removing L_{en} leads to significant performance degradation, highlighting its role in preventing homogeneous cluster assignments. Among the contrastive losses, L_{cl}^2 shows dataset-dependent behavior: its absence minimally affects the Handwritten dataset but severely degrades performance on COIL-20, emphasizing its importance in ensuring robustness across different data characteristics. Additionally, removing L_{cl}^1 causes the model to rely solely on graph embeddings. The performance decline demonstrates the importance of contrasting attribute embeddings with graph embeddings, reducing dependence on the initial graph structure.

4.5.4. Superiority of clustering-oriented mechanism

Unlike sample-based contrastive learning (Xu et al., 2022b), we propose a hierarchical contrastive learning framework comprising: (1) feature-level contrastive learning to separate samples from irrelevant cluster centroids, and (2) cluster-level contrastive learning to align soft cluster assignment. To verify the effectiveness of this clustering-oriented guidance, we design ablation variants by replacing the proposed contrastive components with standard sample-based ones (i.e., treating embeddings of the same sample as positive samples and all other embeddings as negative ones). Specifically, DMCGC-F replaces feature-level contrastive learning, DMCGC-C replaces cluster-level contrastive learning, and DMCGC-B replaces both.

Table 9 summarizes the comparative evaluation of DMCGC against these variants. DMCGC achieves superior mean performance across all datasets. The significant performance gap between DMCGC and DMCGC-B underscores the necessity of clustering-oriented contrastive learning. The results of DMCGC-F and DMCGC-C demonstrate that contrastive learning at both levels is effective, and cluster-level contrastive learning exhibits a greater impact on clustering performance than feature-level contrastive learning.



(a) Caltech-5V (α, β) (b) Caltech-5V (λ, γ) (c) COIL-20 (α, β) (d) COIL-20 (λ, γ)

(e) Caltech-5V (α, β) (f) Caltech-5V (λ, γ) (g) COIL-20 (α, β) (h) COIL-20 (λ, γ)

Fig. 5. ACC and NMI on two datasets with different hyperparameters α, β, λ, and γ.

4.6. Computational efficiency analysis

We compare DMCGC with representative methods on Reuters, which contains 18,758 samples, using an NVIDIA RTX 4070 Laptop GPU. For a coarse comparison, Table 10 reports the complexity of full-data pairwise computation with other model parameters fixed. Under this setting, the compared methods have quadratic cost in N. DMCGC also has $\mathcal{O}(N^2)$ cost when exhaustive neighbor search is used and explicit inversion is replaced by finite-step PPR. Sparse graph storage is adopted for Reuters.

DMCGC requires 414.84 seconds and 1.45 GB of peak GPU memory. Its training time is shorter than those of MFLVC, MAGA, and SSLNMVC, and approximately 18% longer than that of GCFAgg. Its memory usage lies within the range of the compared methods. These results indicate moderate computational overhead and a competitive balance between clustering performance and efficiency on Reuters.

4.7. Parameter sensitivity analysis

In this section, we investigate the impact of the key hyperparameters α, β, λ, and γ, as well as the graph filtering order t, on the clustering performance.

4.7.1. Sensitivity to α, β, λ, and γ

We employ a grid search strategy for pairwise evaluation of α, β, λ, and γ, with the results on Caltech-5V and COIL-20 shown in Fig. 5. The main observations are summarized as follows.

(1) **Stable performance in a low-value regime.** The model consistently achieves favorable performance when the parameters are set to relatively small values, typically below 0.005. Within this range, both ACC and NMI remain relatively high and stable, indicating that DMCGC is insensitive to minor parameter variations within a reasonable range.

(2) **Performance degradation for overly large values.** When the parameters become excessively large, especially above 0.01, the clustering performance generally decreases, since overly large coefficients may disturb the balance among different learning objectives.

(3) **Interaction between α and β.** The results further show that appropriately balancing α and β leads to favorable clustering performance, indicating the complementary effect of different cluster-level alignment terms.

Overall, DMCGC exhibits desirable stability within a relatively broad low-value range, making the effective parameter region easy to identify.

4.7.2. Sensitivity to filtering order t

To examine the influence of the graph filtering order and the potential over-smoothing issue, we vary t in $\{1, 3, 5, 7, 9\}$ on COIL-20 and Caltech-5V while keeping the other settings unchanged. The results are reported in Table 11.

Table 12
Performance under different cluster imbalance ratios on Handwritten.

IR	Method	ACC	NMI	PUR
1	w/o $\mathcal{L}_{en}$	68.70 ± 4.07	84.42 ± 1.84	68.85 ± 3.97
	DMCGC (0.1γ)	88.66 ± 1.21	82.17 ± 1.05	88.66 ± 1.21
	DMCGC	**97.55 ± 0.67**	**94.41 ± 0.94**	**97.55 ± 0.67**
5	w/o $\mathcal{L}_{en}$	72.11 ± 2.19	76.04 ± 1.48	77.30 ± 1.87
	DMCGC (0.1γ)	**74.85 ± 1.86**	**77.14 ± 1.37**	**79.23 ± 1.55**
	DMCGC	67.85 ± 2.57	67.74 ± 1.91	75.02 ± 2.04
10	w/o $\mathcal{L}_{en}$	74.51 ± 2.08	74.59 ± 1.62	80.33 ± 1.63
	DMCGC (0.1γ)	**75.38 ± 1.95**	**75.62 ± 1.55**	**81.14 ± 1.42**
	DMCGC	67.69 ± 2.73	72.80 ± 1.82	79.92 ± 1.82

As shown in Table 11, DMCGC maintains stable performance with moderate filtering orders, while excessively large t gradually degrades both ACC and NMI. This indicates that a moderate order can enlarge the receptive field and capture broader structural information without causing severe over-smoothing, whereas overly high-order filtering may introduce excessive smoothing. Therefore, the small filtering orders adopted in DMCGC provide a suitable balance between global information propagation and discriminative representation preservation.

4.8. Influence of cluster imbalance

To examine the influence of entropy regularization under imbalanced cluster distributions, we construct controlled long-tailed variants of Handwritten with imbalance ratios (IRs) of 1, 5, and 10. Specifically, the number of retained samples for the r-th class is exponentially decreased as $n_r = \lfloor n_{\max} \mathrm{IR}^{-r/(K-1)} \rfloor$, where K is the number of clusters and $\mathrm{IR} = n_{\max}/n_{\min}$. Samples are randomly selected within each class according to the target size. Ground-truth labels are used only to construct these subsets and are not involved in model training.

As shown in Table 12, the default entropy regularization is effective under the balanced setting, but a relatively large weight may introduce bias as the cluster distribution becomes increasingly imbalanced. Reducing the entropy weight to 0.1γ substantially improves performance under IR = 5 and IR = 10, and achieves the best results among the tested variants. This indicates that γ provides a simple and controllable way to reduce the sensitivity of DMCGC to the balanced-cluster assumption.

5. Conclusion

In this paper, we present a multi-view clustering framework, termed DMCGC. Using feature-level contrastive learning, it extracts cluster-centric attribute embeddings for each view, which show improved

adaptability for clustering. Dual-scale graph learning is also proposed to obtain view-shared graph embeddings, and capture both local and global structural relationships among samples. Based on view-specific and view-shared embeddings, cluster-level contrastive supervision aligns the assignments to enforce cross-view consistency, yielding well-separated clusters. Extensive experiments demonstrate that DMCGC effectively leverages graph structures while enhancing clustering-oriented applicability, and rigorous ablation analysis validates the essential role of its core components.

CRediT authorship contribution statement

Jiaxin Zhong: Writing – original draft, Visualization, Validation, Data curation; **Bocheng Wang:** Methodology, Investigation, Formal analysis; **Mulin Chen:** Writing – review & editing, Supervision; **Xuelong Li:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work in this paper.

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